

Post-treatment Fungus Example

Simulation code

```
library(rethinking)
```

```
Loading required package: cmdstanr
```

```
This is cmdstanr version 0.9.0
```

- CmdStanR documentation and vignettes: mc-stan.org/cmdstanr
- CmdStan path: `/Users/thare/.cmdstan/cmdstan-2.36.0`
- CmdStan version: 2.36.0

```
A newer version of CmdStan is available. See ?install_cmdstan() to install it.  
To disable this check set option or environment variable cmdstanr_no_ver_check=TRUE.
```

```
Loading required package: posterior
```

```
This is posterior version 1.6.1
```

```
Attaching package: 'posterior'
```

```
The following objects are masked from 'package:stats':
```

```
mad, sd, var
```

The following objects are masked from 'package:base':

`%in%`, `match`

Loading required package: parallel

rethinking (Version 2.42)

Attaching package: 'rethinking'

The following object is masked from 'package:stats':

`rstudent`

```
set.seed(71)

#number of plants
N <- 100

#simulate initial heights
h0 <- rnorm(N,10,2)

#assign treatments and simulate fungus and growth
treatment <- rep( 0:1, each=N/2)
fungus <- rbinom( N, size=1, prob=0.5 - treatment*0.4)
h1 <- h0 + rnorm(N, 5 - 3*fungus)

#compose a clean data frame
d <- data.frame( h0=h0, h1=h1, treatment=treatment, fungus=fungus)
```

Data summary

```
precis(d)
```

	mean	sd	5.5%	94.5%	histogram
h0	9.95978	2.1011623	6.570328	13.07874	
h1	14.39920	2.6880870	10.618002	17.93369	
treatment	0.50000	0.5025189	0.000000	1.00000	
fungus	0.23000	0.4229526	0.000000	1.00000	

```
table(d[, c("treatment", "fungus")])
```

```
      fungus
treatment 0  1
      0 31 19
      1 46  4
```

Model code

```
m6.7 <- quap(
  alist( h1 ~ dnorm( mu, sigma),
        mu <- h0 * p,
        p <- a + bt*treatment + bf*fungus,
        a ~ dlnorm( 0, 0.2),
        bt ~ dnorm( 0, 0.5),
        bf ~ dnorm( 0, 0.5),
        sigma ~ dexp( 1) ),
  data=d)
```

Simulate from priors

```
sim_p <- function(treat = 0, infect = 0) {
  a <- rlnorm(1, 0, 0.2)
  bt <- rnorm(1, 0, 0.5)
  bf <- rnorm(1, 0, 0.5)
  p <- a + bt*treat + bf*infect

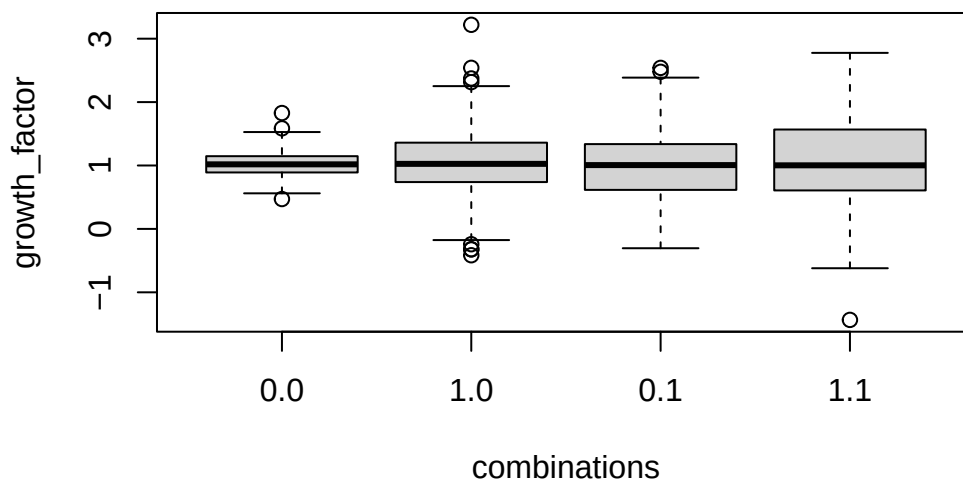
  return(p)
}

# Random treatment and fungus infection combinations
treat <- rbinom(1000, 1, 0.5)
infect <- rbinom(1000, 1, 0.5)
rand_data <- data.frame(treat = treat,
                       infect=infect)
combinations <- interaction(treat, infect)
```

```
# Apply sim_p to each row of rand_data
growth_factor <- mapply(sim_p, treat = rand_data$treat, infect = rand_data$infect)
```

Simulate from priors

```
boxplot(growth_factor ~ combinations)
```



Fitting the model

```
m6.7 <- quap(
  alist( h1 ~ dnorm( mu, sigma),
        mu <- h0 * p,
        p <- a + bt*treatment + bf*fungus,
        a ~ dlnorm( 0, 0.2),
        bt ~ dnorm( 0, 0.5),
        bf ~ dnorm( 0, 0.5),
        sigma ~ dexp( 1) ),
  data=d)
```

```
precis(m6.7, prob = 0.91) #change the interval from 89 to 91 just for fun
```

	mean	sd	4.5%	95.5%
a	1.481392148	0.02451072	1.43983673	1.52294756
bt	0.002410437	0.02986968	-0.04823055	0.05305142
bf	-0.266718019	0.03654776	-0.32868102	-0.20475502
sigma	1.408799050	0.09862098	1.24159726	1.57600084

The treatment is not associated with growth?

- The marginal posterior for bt, the effect of treatment, is solidly zero, with a tight interval.
- We simulated the data such that treatment matters, what happened here?

Blocked by consequence

The problem is that fungus is mostly a consequence of treatment.

Fungus is a post-treatment variable.

Blocked by consequence

When we control for fungus, the model is implicitly answering the question: Once we already know whether or not a plant developed fungus, does soil treatment matter?

The answer is “no,” because soil treatment has its effects on growth through reducing fungus.

A better model for our real question

```
m6.8 <- quap( alist(
  h1 ~ dnorm( mu, sigma),
  mu <- h0 * p,
  p <- a + bt*treatment,
  a ~ dlnorm( 0, 0.2),
  bt ~ dnorm( 0, 0.5),
  sigma ~ dexp( 1) ),
  data=d)
```

```
precis(m6.8)
```

	mean	sd	5.5%	94.5%
a	1.38035501	0.02517629	1.34011844	1.4205916
bt	0.08499795	0.03429818	0.03018284	0.1398131
sigma	1.74636818	0.12192450	1.55150928	1.9412271

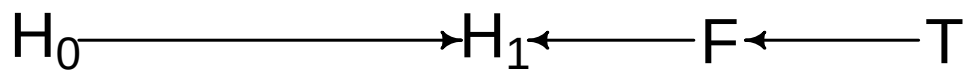
Using dagitty package

```
library(dagitty)

plant_dag <- dagitty( "dag {
    H_0 -> H_1
    F -> H_1
    T -> F }")

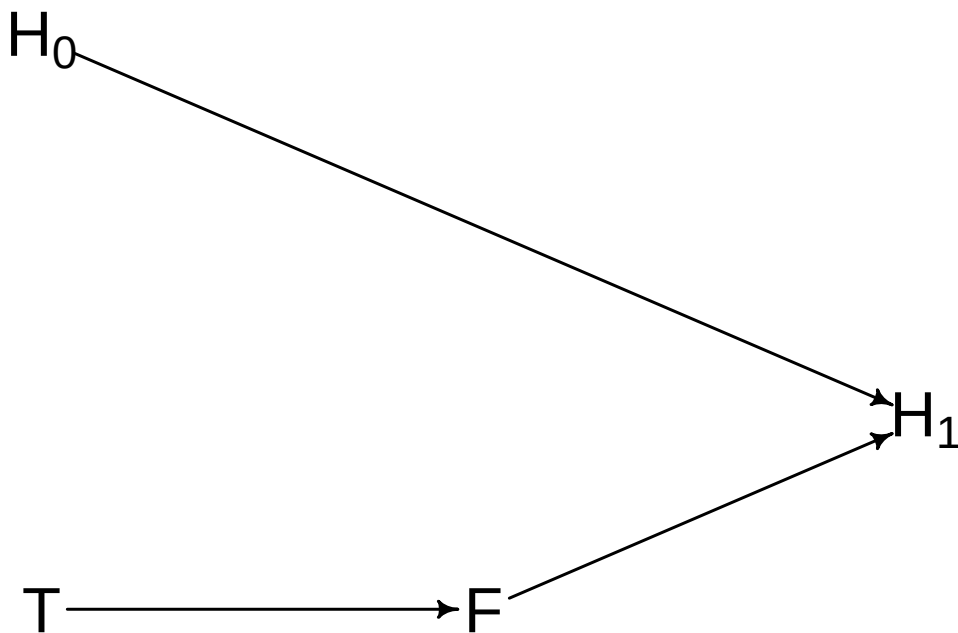
coordinates( plant_dag) <- list( x=c(H_0=0,T=2,F=1.5,H_1=1),
                                y=c(H_0=0,T=0,F=0,H_1=0))

drawdag( plant_dag, cex = 2)
```



Using dagitty package

```
1 plant_dag <- dagitty( "dag {  
2     H_0 -> H_1  
3     F -> H_1  
4     T -> F  
5     H_0 [ancestor of exposure]  
6     H_1 [outcome]  
7     T [exposure]  
8     F [covariate]  
9     }"  
10  
11 coordinates( plant_dag) <- list( x=c(H_0=0,T=0,F=0.5,H_1=1),  
12                                   y=c(H_0=0,T=3,F=3,H_1=2))  
13 drawdag( plant_dag, cex = 2)
```



Using dagitty package

```
impliedConditionalIndependencies(plant_dag)
```

F _||_ H_0
H_0 _||_ T
H_1 _||_ T | F